

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0509

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks:50

Annexure A Short Answer Test

Code of Conduct:

I B.Sivapavathi (192525194) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Sivapavathi

Short Answer Test Questions

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.
2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.
3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.
 5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.
 6. Explain how image formation models help in improving image quality in applications such as autonomous driving.
 7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.
 8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.
 9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.
 10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.
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- Sampling decides the number of Pixels used to represent an image.

- Low Sampling causes blur and loss of details.

- Increase the Sampling rate (higher resolution) and use anti-aliasing filters to improve image quality.

2)

- In low light, the sensor captures less light, causing dark and noisy images.

- Better sensors, proper exposure, and good lighting improve image quality.

3)

- Aliasing occurs due to low sampling or improper quantization.

- It causes jagged edges and distortion.

- Use higher sampling resolution and anti-aliasing filters to reduce it.

4) Pixel Resolution

1) Number of Pixels in an image.

2) Affects image sharpness.

Ex:- 1024 x 768

Intensity Resolution

1) Number of gray/color levels.

2) Affects brightness and contrast.

Ex:- 8-bit = 256 levels.

5) 1) High level computer-vision.

2) It detects, recognizes, and tracks people or objects in real time to make decisions.

6) 1) They explain how cameras capture images.

2) They improve image clarity, depth estimation, and object detection.

3) This helps detect roads, vehicles, and pedestrians accurately.

7) 1) Quantization Converts continuous intensity values into discrete digital values.

2) High quantization levels preserve more image details.

3) Lower levels cause loss of detail and banding.

8) 1) Noise is introduced by sensors, poor lighting, and electronic interference.

2) It can be reduced using better sensors, proper lighting, and noise filtering techniques.

9) 1) Image Acquisition - capture face image.

2) Low-level vision - Remove noise and enhance image.

3) Mid-level vision - Detect and extract facial features.

4) High-level vision - Recognize the person and make a decision.

10)

- 1) Low sampling produces low-resolution images, making objects unclear.
- 2) Poor image formation causes blur and distortion.
- 3) High-resolution sampling and proper image formation improve object detection.